

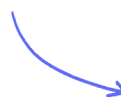
IGCSE • Cambridge (CIE) • Biology

 1 hour  16 questions

Target Test

Created 30/05/26

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Total Marks

/70

1 A potato tissue sample has an initial mass of 5.0 g.

It is immersed in an unknown solution.

Its mass after 1 hour is 4.2 g.

Which statement is correct?

- A.** The solution had a lower water potential than the plant tissue.
- B.** The solution had a higher water potential than the plant tissue.
- C.** The plant tissue had no cell membrane.
- D.** The solution was distilled water.

(1 mark)

2 Which statement describes the net movement of particles during diffusion?

- A.** From higher to lower concentration down a concentration gradient
- B.** From higher to lower concentration against a concentration gradient
- C.** From lower to higher concentration down a concentration gradient
- D.** From lower to higher concentration against a concentration gradient

(1 mark)

3 (a) Water is important for living organisms as many substances are able to dissolve in it.

Give **two** examples of the importance of water as a solvent in living organisms.

(2 marks)

(b) Fig. 1 shows water and a sugar solution separated by a partially permeable membrane.

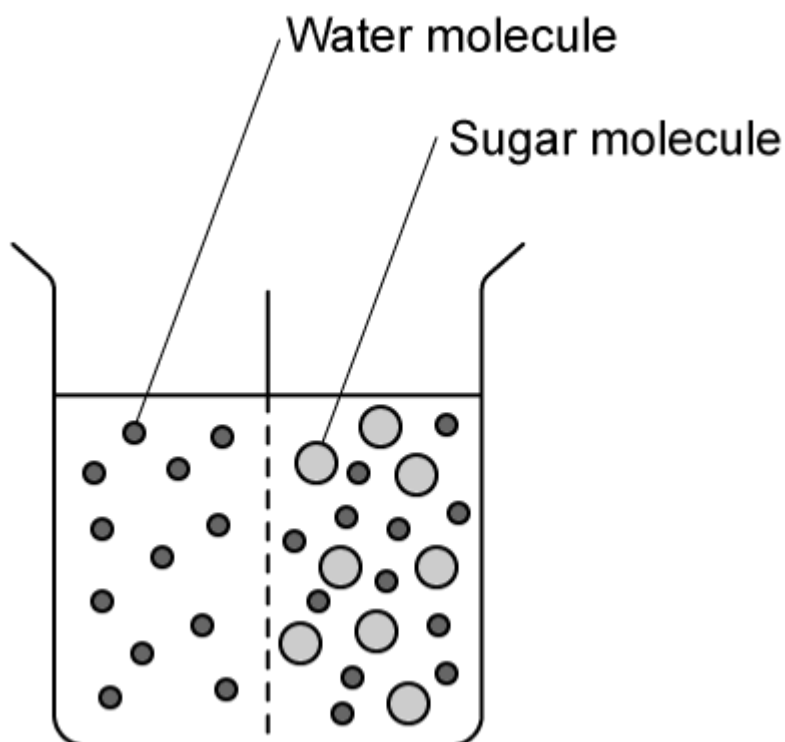


Fig. 1

State the overall direction in which water molecules will move.

..... (1 mark)

(c) Osmosis occurred in Fig. 1.

Define the term 'osmosis'.

..... (2 marks)

(d) A second beaker, similar to the one from Fig. 1, was set up and placed in a water bath that was 10°C lower than the beaker from Fig. 1.

Explain how the movement of the water molecules would be affected by this decrease in temperature.

4 Three statements about the role of water in mammals are given below:

1. Water forms a major part of the cytoplasm, allowing substances to dissolve and participate in metabolic reactions within the cell
2. Substances, such as urea and excess salts, are able to dissolve in water making them easier to remove from the body in urine
3. Allows substances to be transported around the body in the blood

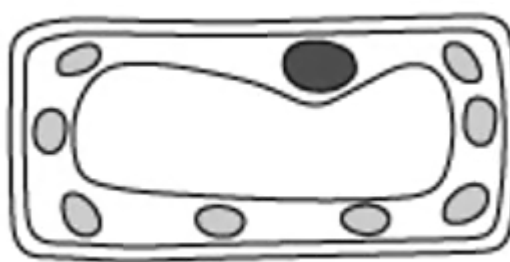
Which of the statements above are true?

- A. 1 & 2 only
- B. 2 & 3 only
- C. All of the statements
- D. 1 & 3 only

(1 mark)

5 Extended only

The image below shows a leaf palisade cell:

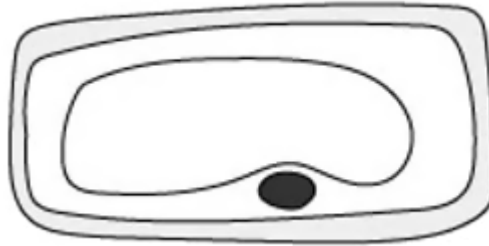


What would the correct appearance of the leaf palisade cell be after it has been placed in a highly concentrated solution of sugar after an hour?

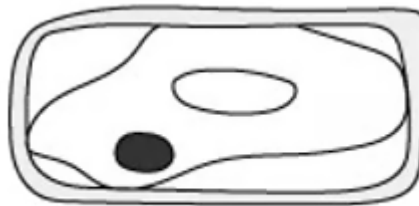
A.



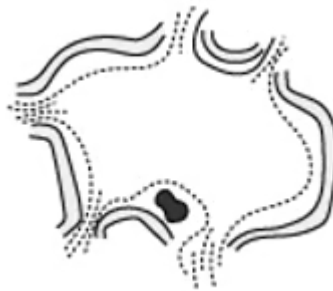
B.



C.



D.



(1 mark)

6 What are the features of active transport?

	occurs through a cell membrane	particles move from a higher concentration to a lower concentration	uses energy from respiration
A	Yes	Yes	Yes
B	Yes	Yes	No
C	Yes	No	Yes
D	No	Yes	Yes

(1 mark)

- 7** A cylinder of potato tissue was placed in a beaker of very salty water. After one hour the mass of the potato cylinder had decreased.

Why did this happen?

- A.** Salt entered the plant tissue by active transport.
- B.** Salt left the plant tissue by osmosis.
- C.** Water entered the plant tissue by active transport.
- D.** Water left the plant tissue by osmosis.

(1 mark)

8 A plant cell is placed in a concentrated salt solution.

Which of the following correctly describes the events that occur after this?

- A.** Water molecules will move out of the plant cell via osmosis into the salt solution. Eventually the cell will become plasmolysed.
- B.** Water molecules will move out of the plant cell via active transport into the salt solution. Eventually the cell will plasmolyse.
- C.** Water molecules will move into the plant cell via osmosis from the salt solution. Eventually the cell will become turgid.
- D.** Water molecules will move out of the plant cell via osmosis into the salt solution. Eventually the cell will become flaccid.

(1 mark)

- 9 (a)** Scientists grew two groups of carrot seedlings in a solution containing magnesium ions, required for the seedlings to grow. The magnesium ions in the solution were radioactively labelled.

The scientists bubbled oxygen through the solution of one group and pure nitrogen gas through the solution of the other group. Nitrogen gas will remove all dissolved oxygen from the solution.

The level of radioactive magnesium ions was measured at the start of the experiment and again, every 30 minutes for five hours. The scientists subtracted the level of magnesium ions in the solution from the starting level as a way of determining the uptake of magnesium ions by the plant. Table 1 shows the scientists' results.

Table 1

time / minutes	total level of magnesium ions absorbed / arbitrary units	
	with oxygen	without oxygen
0	0	0
30	190	132
60	224	148
90	237	163
120	263	174
150	289	189
180	320	206
210	342	225
240	368	238
270	385	246
300	412	257

(i)

Plot a line graph of the results shown in Table 1.

[4]

(ii)

State the solution the carrot seedlings absorbed more magnesium ions from.

[1]

(5 marks)

(b) (i)

Identify the independent **and** dependent variables in the investigation.

[2]

(ii)

Identify a variable that needs to be kept constant (control variable) throughout the investigation.

[1]

(iii)

Calculate the percentage increase of magnesium ions absorbed with oxygen between 120 minutes and 300 minutes.

[3]

(iv)

Calculate the range for the level of magnesium ions absorbed without oxygen from the beginning to the end of the experiment.

[1]

(7 marks)

(c) Describe how the scientists could make the results reliable.

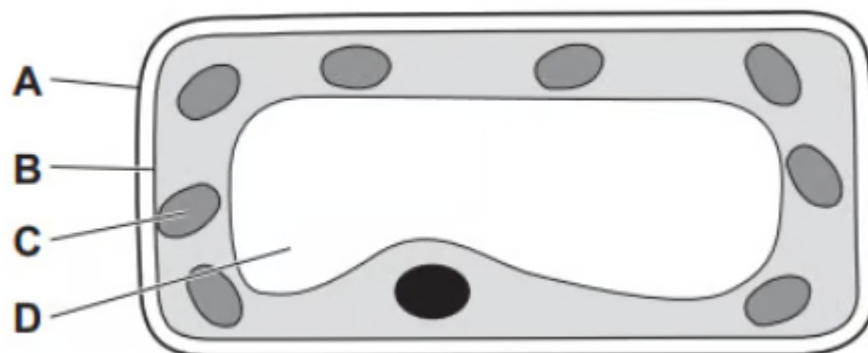
(2 marks)

(d) Explain why the scientists took data over a five hour period.

(2 marks)

10 The diagram shows a section through a mesophyll cell of a leaf.

Which part is partially permeable?



11 (a) Fig. 1.1 is a diagram of a plant cell.

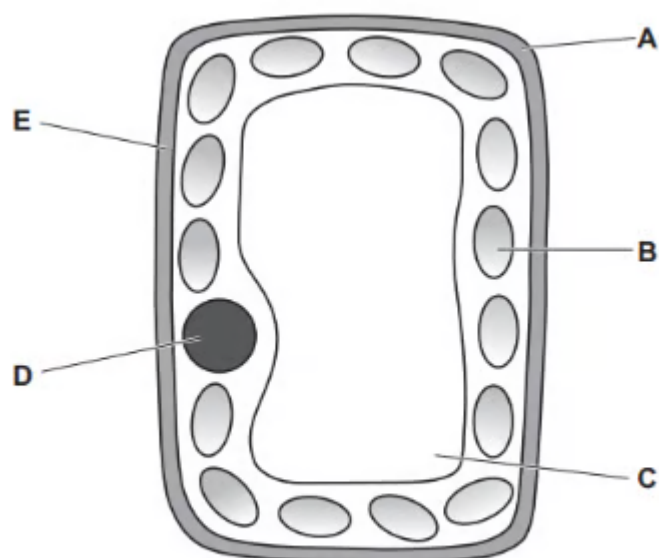


Fig. 1.1

The boxes on the left contain the letters that identify some parts of the plant cell in Fig. 1.1.

The boxes on the right show the functions of some parts of a cell.

Draw five lines to link each letter to its correct function.

letter from Fig. 1.1	function
A	contains the genetic material
B	controls which substances enter and leave the cell
C	filled with sap to support the cell
D	strengthens the cell
E	transports nerve impulses
	where photosynthesis occurs

(5 marks)

(b) State the names of **two** different types of plant cell.

.....
..... (2 marks)

(c) A student used a potato as a source of plant tissue. The student cut six cylinder-shaped pieces from the potato.

Each potato cylinder had the same diameter. Each potato cylinder was immersed in either water or one of five different concentrations of sugar solution.

The student measured the length of the potato cylinders before immersion and after being immersed for 30 minutes.

Table 1.1 shows the results.

Table 1.1

concentration of sugar solution /mol per dm ³	length of potato cylinder before immersion /mm	length of potato cylinder after immersion /mm	change in length of potato cylinder /mm
0.0	49.5	52.0	+ 2.5
0.2	50.0	52.0	+ 2.0
0.4	50.5	51.5	+ 1.0
0.6	50.5	51.0	+ 0.5
0.8	50.0	49.0	
1.0	50.0	48.5	- 1.5

(i) Use the information in Table 1.1 to calculate the change in length of the potato cylinder immersed in the 0.8mol per dm³ sugar solution.

..... mm [1]

(ii) Use the information in Table 1.1 to calculate the percentage increase in length of the potato cylinder immersed in the 0.2mol per dm³ sugar solution

..... % [2]

.....

.....

..... (3 marks)

(d) State the name of the process that causes water to enter or leave the potato cells.

..... (1 mark)

- (e) Describe the expected change in appearance of a potato cell that was immersed in a 1.0mol per dm³ sugar solution.

..... (1 mark)

12 By which process do oxygen and carbon dioxide move between cells and capillaries?

- A. breathing
- B. diffusion
- C. excretion
- D. respiration

(1 mark)

13 (a) A group of scientists investigated the rates of absorption of different sugars using two pieces of the intestine.

One of these pieces were treated with cyanide, which is a toxin that prevents cellular respiration from occurring.

The results of the investigation are shown in Table 1.

Table 1

sugar	absorption rate / arbitrary units	
	healthy intestine	intestine treated with cyanide
A	121	62
B	47	45
C	68	67
D	89	35

Suggest the method by which sugar **A**, **B** and **C** are absorbed by the intestine.

(3 marks)

- (b)** The scientists concluded that sugar **D** is absorbed by active transport.

Explain how they came to this conclusion by using evidence from Table 1.

(3 marks)

- (c)** One of the scientists states that 'All four of the sugars investigated could be absorbed by diffusion'.

Determine whether this statement is correct by using the information in Table 1 and your knowledge of diffusion.

(2 marks)

(d) Extended only

The wall of the intestine contains smooth muscle cells which convert glucose into glycogen. Glycogen is insoluble in water which makes it very suitable as an energy storage molecule.

Using your knowledge of osmosis, suggest an explanation as to why it is important for glucose to be stored as glycogen in muscle cells.

14 Which of the following statements would be correct for both diffusion and osmosis?

	does not require energy from the cell	molecules must be separated by a partially permeable membrane	molecules move from a dilute to a more concentrated solution
A	✓	✓	X
B	X	X	✓
C	✓	X	X
D	✓	X	✓

(1 mark)

15 How do carbon dioxide and oxygen move into and out of a mesophyll cell?

- A.** Active transport
- B.** Diffusion
- C.** Osmosis
- D.** Transpiration

(1 mark)

16 (a) Extended only

Fig. 1 shows two plant cells, each placed in a solution with a different concentration.

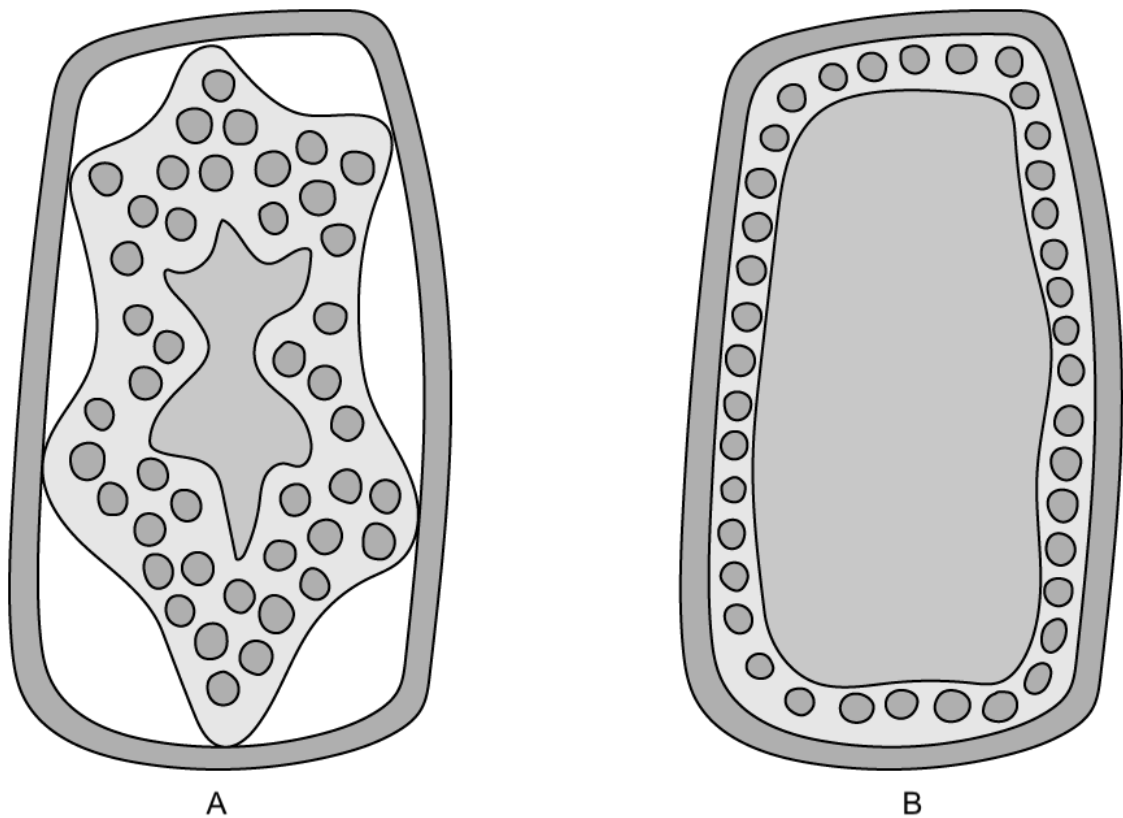


Fig. 1

Which of the plant cells above (**A** or **B**) was placed in a solution that was more dilute than the cell contents?

(1 mark)

(b) Extended only

Explain what has happened to the plant cell you identified in part (a).

(4 marks)

(c) Extended only

When placed in different solutions, animal cells will also change.

Fig. 2 below shows a red blood cell.

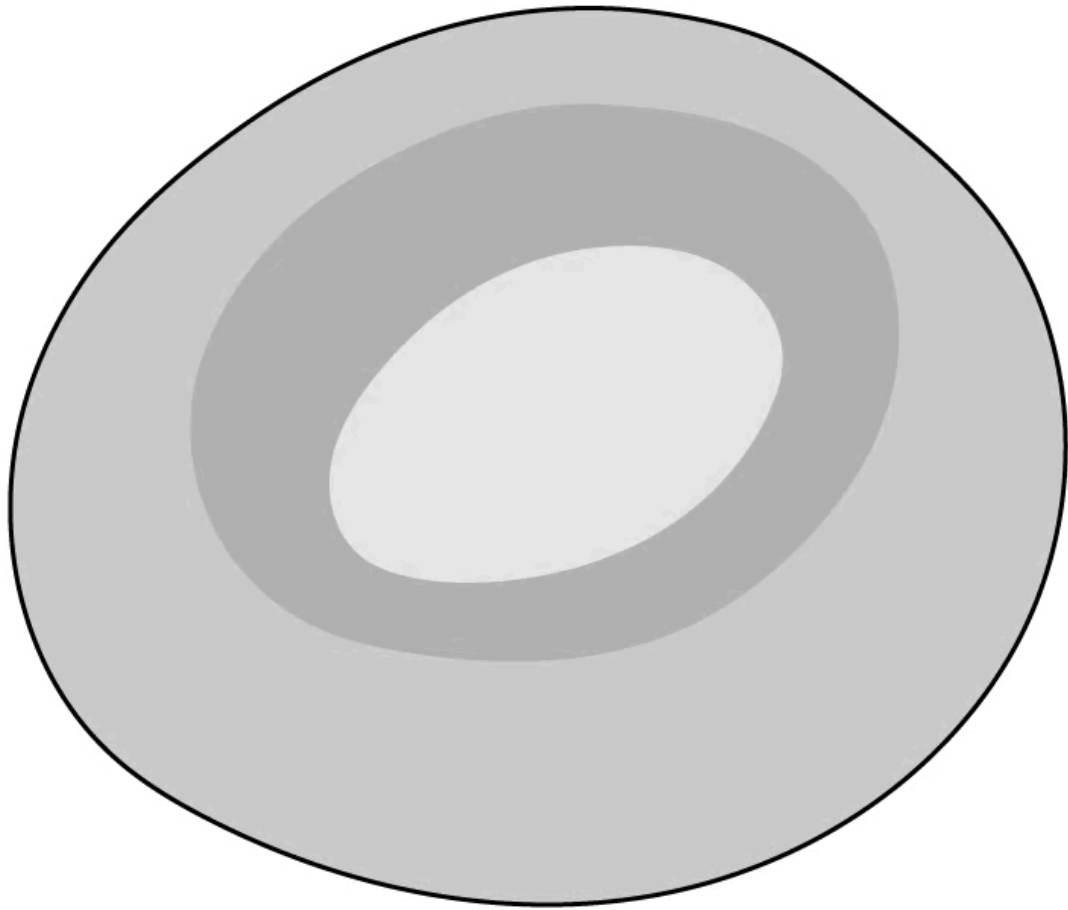


Fig. 2

(i)

Describe what would happen if a red blood cell was placed in a solution that was more dilute than the cell contents.

[2]

(ii)

Explain why the description at part (i) would be different from what would happen to a plant cell.

(4 marks)

- (d) Hereditary spherocytosis is a rare inherited condition that produces red blood cells with an abnormal spherical shape, as shown in Fig. 3.

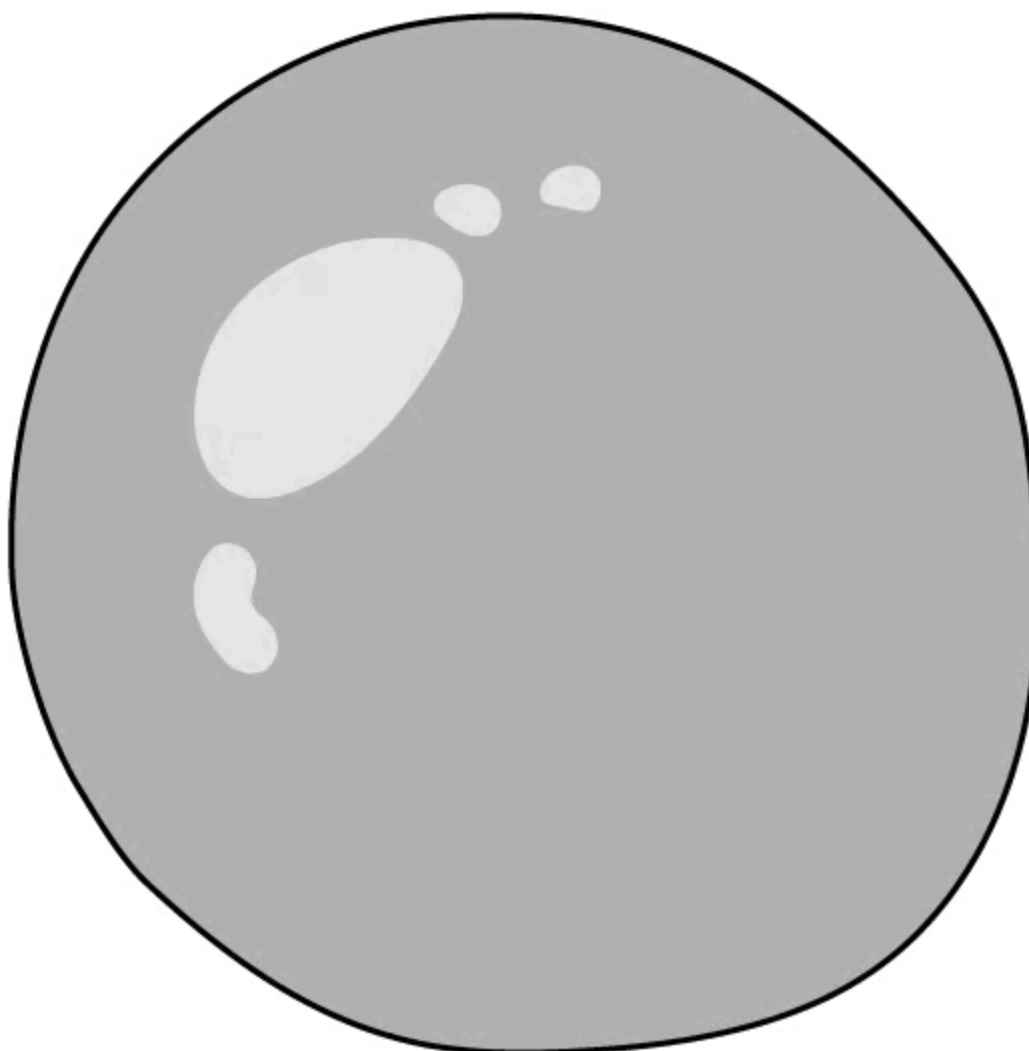


Fig. 3

Suggest the possible impact of this condition on a sufferer.

(3 marks)